Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Growing Patterns

Name _____ Date _____ No calculator Version 1

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KEY WORDS pattern *(tauira)* · term *(tūnga)* · rule *(ture)* · increase *(pikinga)* · decrease *(hekenga)* · input · output

A Adding and subtracting patterns ●○○

Remember: Find what is added or subtracted each time. Check that it is the same between every pair of numbers.

1. Write the next two terms of each pattern.

a) 3, 7, 11, 15, ___, ___

b) 5, 11, 17, 23, ___, ___

c) 20, 17, 14, 11, ___, ___

d) 2, 11, 20, 29, ___, ___

e) 50, 43, 36, 29, ___, ___

f) 8, 19, 30, 41, ___, ___

g) 100, 85, 70, 55, ___, ___

h) 125, 150, 175, 200, ___, ___

B Decimals and fractions ●●○

2. Write the next two terms of each pattern.

a) 0.5, 1, 1.5, 2, ___, ___

b) 0.2, 0.4, 0.6, 0.8, ___, ___

c) 1.5, 1.75, 2, 2.25, ___, ___

d) 3, 2.5, 2, 1.5, ___, ___

e) $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, 1, ___, ___

f) $\frac{1}{3}$, $\frac{2}{3}$, 1, $1\frac{1}{3}$, ___, ___

C Multiplying patterns ●●○

Remember: Some patterns grow by multiplying. In 2, 4, 8, 16, … each number is double the one before.

3. Write the next two terms of each pattern.

a) 1, 2, 4, 8, ___, ___

b) 1, 3, 9, 27, ___, ___

c) 5, 10, 20, 40, ___, ___

d) 3, 30, 300, 3,000, ___, ___

e) 96, 48, 24, 12, ___, ___

f) 2, 6, 18, 54, ___, ___

D Describing a rule ●●○

4. Describe the rule for each pattern in words.

a) 4, 7, 10, 13, 16

b) 30, 25, 20, 15, 10

c) 2, 4, 8, 16, 32

d) 7, 7.5, 8, 8.5, 9

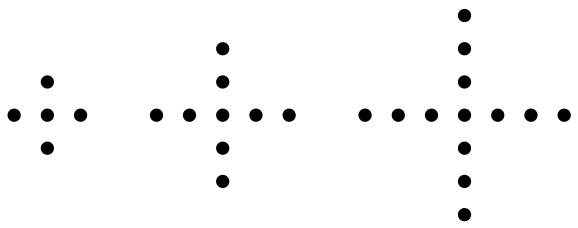
e) 1, 10, 100, 1,000, 10,000

f) 64, 32, 16, 8, 4

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E Patterns from diagrams ●●●

5. Dots are drawn in the shape of a plus sign.



Pattern 1 **Pattern 2** **Pattern 3**

a) Complete the table.

Pattern	1	2	3	4	5
Number of dots					

b) How many dots are added each time?

c) How many dots are there in pattern 8?

F Predicting further terms ●●●

6. Answer each question.

a) The pattern 5, 8, 11, 14, ... carries on in the same way. What is the 10th term?

b) The pattern 4, 10, 16, 22, ... carries on in the same way. What is the 8th term?

c) The pattern 50, 46, 42, 38, ... carries on in the same way. What is the 10th term?

G Tables of values from a rule ●●●

> **Remember:** Put each input into the rule to find its output. For the rule "multiply the input by 2, then add 1", the input 3 gives $3 \times 2 + 1 = 7$.

7. Use the rule to complete each table.

a) Rule: add 5

Input	1	2	3	4	5
Output					

b) Rule: multiply the input by 3

Input	1	2	3	4	5
Output					

c) Rule: multiply the input by 2, then add 1

Input	1	2	3	4	5
Output					

d) Rule: multiply the input by 3, then subtract 1

Input	2	4	6	8	10
Output					

★ Extension

8. The pattern 3, 6, 12, 24, ... is multiplied by 2 each time. What is the first term that is larger than 100?

9. Look at the pattern 1, 4, 9, 16, ... Write the next two terms. What is added each time?

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Growing Patterns

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS pattern *(tauira)* · term *(tūnga)* · rule *(ture)* · increase *(pikinga)* · decrease *(hekenga)* · input · output

A Adding and subtracting patterns ●○○

Remember: Find what is added or subtracted each time. Check that it is the same between every pair of numbers.

1. Write the next two terms of each pattern.

a) 2, 7, 12, 17, ___, ___

b) 4, 11, 18, 25, ___, ___

c) 25, 21, 17, 13, ___, ___

d) 3, 11, 19, 27, ___, ___

e) 60, 51, 42, 33, ___, ___

f) 7, 19, 31, 43, ___, ___

g) 90, 75, 60, 45, ___, ___

h) 150, 175, 200, 225, ___, ___

B Decimals and fractions ●●○

2. Write the next two terms of each pattern.

a) 1.5, 2, 2.5, 3, ___, ___

b) 0.3, 0.6, 0.9, 1.2, ___, ___

c) 2.5, 2.75, 3, 3.25, ___, ___

d) 4, 3.5, 3, 2.5, ___, ___

e) $\frac{1}{2}$, 1, $1\frac{1}{2}$, 2, ___, ___

f) $\frac{1}{5}$, $\frac{2}{5}$, $\frac{3}{5}$, $\frac{4}{5}$, ___, ___

C Multiplying patterns ●●○

Remember: Some patterns grow by multiplying. In 2, 4, 8, 16, … each number is double the one before.

3. Write the next two terms of each pattern.

a) 3, 6, 12, 24, ___, ___

b) 2, 6, 18, 54, ___, ___

c) 5, 10, 20, 40, ___, ___

d) 4, 40, 400, 4,000, ___, ___

e) 160, 80, 40, 20, ___, ___

f) 1, 4, 16, 64, ___, ___

D Describing a rule ●●○

4. Describe the rule for each pattern in words.

a) 5, 9, 13, 17, 21

b) 40, 34, 28, 22, 16

c) 1, 3, 9, 27, 81

d) 3, 3.5, 4, 4.5, 5

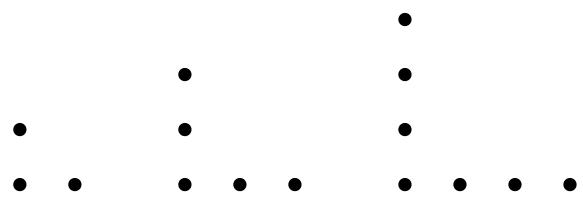
e) 2, 20, 200, 2,000, 20,000

f) 80, 40, 20, 10, 5

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E Patterns from diagrams ●●●

5. Dots are drawn in the shape of a letter L.



Pattern 1 **Pattern 2** **Pattern 3**

a) Complete the table.

Pattern	1	2	3	4	5
Number of dots					

b) How many dots are added each time?

c) How many dots are there in pattern 8?

F Predicting further terms ●●●

6. Answer each question.

a) The pattern 6, 10, 14, 18, ... carries on in the same way. What is the 10th term?

b) The pattern 3, 10, 17, 24, ... carries on in the same way. What is the 8th term?

c) The pattern 60, 55, 50, 45, ... carries on in the same way. What is the 10th term?

G Tables of values from a rule ●●●

> **Remember:** Put each input into the rule to find its output. For the rule "multiply the input by 2, then add 1", the input 3 gives $3 \times 2 + 1 = 7$.

7. Use the rule to complete each table.

a) Rule: add 7

Input	1	2	3	4	5
Output					

b) Rule: multiply the input by 4

Input	1	2	3	4	5
Output					

c) Rule: multiply the input by 2, then add 3

Input	1	2	3	4	5
Output					

d) Rule: multiply the input by 4, then subtract 1

Input	2	4	6	8	10
Output					

★ Extension

8. The pattern 2, 6, 18, 54, ... is multiplied by 3 each time. What is the first term that is larger than 200?

9. Look at the pattern 1, 4, 9, 16, ... Write the next two terms. What is added each time?

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